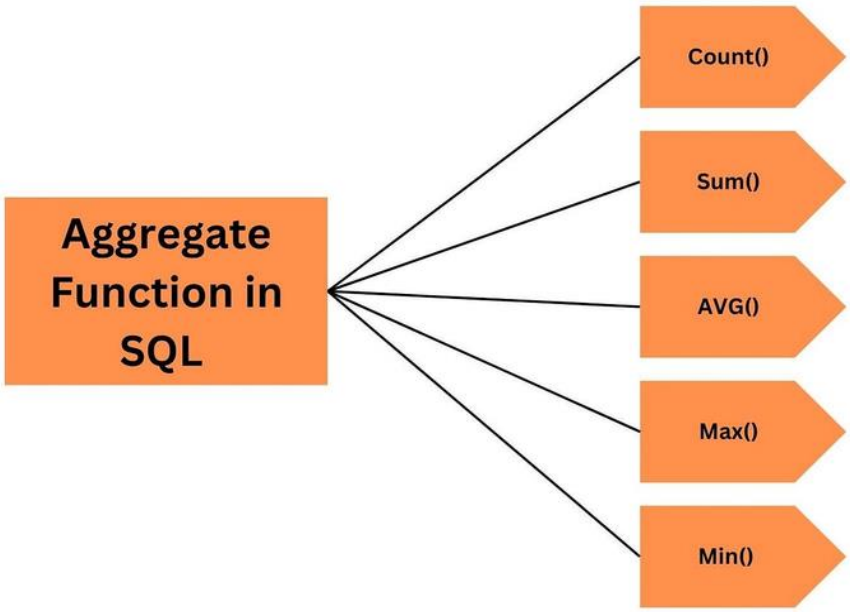
DBMS – SALES AND CUSTOMER ANALYTICS SYSTEM
SWIGGY INSTAMART E-COMMERCE DATABASE MANAGEMENT SYSTEM

Objectives

1. Apply COUNT(), SUM(), AVG(), MIN(), and MAX().
2. Generate total sales reports.
3. Find top customers based on purchase amount.
4. Identify best-selling products.
5. Perform category-wise sales analysis.

1. Aggregate Functions

Aggregate functions are SQL functions used to perform calculations on a group of rows and return a single result.



1.1 COUNT()

Find the total number of orders:

```
SQL> SELECT COUNT(Order_ID) AS Total_Orders
  2  FROM Orders;

TOTAL_ORDERS
------------
          10
```

1.2 SUM()

Find the total sales amount:

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM Orders;

TOTAL_SALES
-----------
       6835
```

1.3 AVG()

Find the average order amount:

```
SQL> SELECT AVG(Total_Amount) AS Average_Order_Amount
  2  FROM Orders;

AVERAGE_ORDER_AMOUNT
--------------------
              683.5
```

1.4 MIN()

Find the lowest order amount:

```
SQL> SELECT MIN(Total_Amount) AS Minimum_Order
  2  FROM Orders;

MINIMUM_ORDER
-------------
          320
```

1.5 MAX()

Find the highest order amount:

```
SQL> SELECT MAX(Total_Amount) AS Maximum_Order
  2  FROM Orders;

MAXIMUM_ORDER
-------------
         1050
```

2. Generate Total Sales Reports

A sales report can be generated using the SUM() function on the Total_Amount column.

```
SQL> SELECT
  2      COUNT(Order_ID) AS Total_Orders,
  3      SUM(Total_Amount) AS Total_Sales,
  4      AVG(Total_Amount) AS Average_Order_Value,
  5      MIN(Total_Amount) AS Minimum_Order,
  6      MAX(Total_Amount) AS Maximum_Order
  7  FROM Orders;

TOTAL_ORDERS TOTAL_SALES AVERAGE_ORDER_VALUE MINIMUM_ORDER MAXIMUM_ORDER
------------ ----------- ------------------ ------------- -------------
          10        6835              683.5           320          1050
```

2.1 Customer-wise Sales Report

To find how much each customer has purchased:

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  5  FROM Customer C
  6  JOIN Orders O
  7      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------
        105
Vikram Singh
          1970

        102
Priya Kumar
          1725

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
```

3. Find Top Customers Based on Purchase Amount

The SUM() function can be combined with GROUP BY and ORDER BY to identify customers with higher purchase amounts.

```
SQL> SELECT * FROM (
  2        SELECT
  3          C.CUSTOMER_ID,
  4          C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  5          SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  6      FROM Customer C
  7      JOIN Orders O
  8      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  9      GROUP BY
 10          C.CUSTOMER_ID,
 11          C.FIRST_NAME,
 12          C.LAST_NAME
 13      ORDER BY TOTAL_PURCHASE DESC
 14  )
 15  WHERE ROWNUM <= 5;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------
         105
Vikram Singh
          1970

         102
Priya Kumar
```

4. Identify Best-Selling Products

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OI.QUANTITY) AS TOTAL_QUANTITY_SOLD
  5  FROM Product P
  6  JOIN Order_Item OI
  7  ON P.PRODUCT_ID = OI.PRODUCT_ID
  8  GROUP BY
  9      P.PRODUCT_ID,
 10      P.PRODUCT_NAME
 11  ORDER BY TOTAL_QUANTITY_SOLD DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
         2
Bread
                 6

         1
Milk
                 5

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------

         3
Apple
                 5

         4
Biscuits
```

4.1 Product Sales Amount

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OI.QUANTITY) AS TOTAL_QUANTITY_SOLD,
  5      SUM(OI.QUANTITY * OI.UNIT_PRICE) AS TOTAL_SALES
  6  FROM Product P
  7  JOIN Order_Item OI
  8  ON P.PRODUCT_ID = OI.PRODUCT_ID
  9  GROUP BY
 10      P.PRODUCT_ID,
 11      P.PRODUCT_NAME
 12  ORDER BY TOTAL_SALES DESC;

PRODUCT_ID
---------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
         4
Biscuits
                    5        1000

         5
Fruit Juice
                    3         960

PRODUCT_ID
---------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------

         2
Bread
                    6         900

         1
```

5. Category-wise Sales Analysis

Category-wise analysis helps determine the sales generated by each product category.

```
SQL> SELECT
  2      C.CATEGORY_ID,
  3      C.CATEGORY_NAME,
  4      SUM(OI.QUANTITY) AS TOTAL_QUANTITY_SOLD,
  5      SUM(OI.QUANTITY * OI.UNIT_PRICE) AS TOTAL_SALES
  6  FROM Category C
  7  JOIN Product P
  8  ON C.CATEGORY_ID = P.CATEGORY_ID
  9  JOIN Order_Item OI
 10  ON P.PRODUCT_ID = OI.PRODUCT_ID
 11  GROUP BY
 12      C.CATEGORY_ID,
 13      C.CATEGORY_NAME
 14  ORDER BY TOTAL_SALES DESC;

CATEGORY_ID CATEGORY_NAME
----------- --------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
        104 Snacks
                 11        1900

        103 Beverages
                  3         960

        105 Dairy
                  5         780


CATEGORY_ID CATEGORY_NAME
----------- --------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
        101 Fruits
                  5         400
```

6. Combined Business Sales Report

A complete sales report can combine customer, order, product, and category information

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      O.ORDER_ID,
  5      P.PRODUCT_ID,
  6      P.PRODUCT_NAME,
  7      CAT.CATEGORY_NAME,
  8      OI.QUANTITY,
  9      OI.UNIT_PRICE,
 10      (OI.QUANTITY * OI.UNIT_PRICE) AS SUBTOTAL
 11  FROM Customer C
 12  JOIN Orders O
 13  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 14  JOIN Order_Item OI
 15  ON O.ORDER_ID = OI.ORDER_ID
 16  JOIN Product P
 17  ON OI.PRODUCT_ID = P.PRODUCT_ID
 18  JOIN Category CAT
 19  ON P.CATEGORY_ID = CAT.CATEGORY_ID
 20  ORDER BY O.ORDER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
       101

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
```

```
---------------------------------------------- ---------- ----------
  SUBTOTAL
----------
       101

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
---------------------------------------------- ---------- ----------
  SUBTOTAL
----------
Rahul Sharma

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
---------------------------------------------- ---------- ----------
  SUBTOTAL
----------
      1001          1

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID PRODUCT_ID
---------- ----------
```